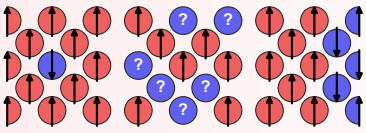
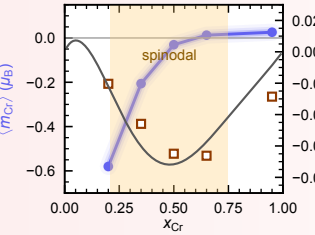
FRUSTRATE → NEST → STIFFEN → DETECT

(a) atoms & moments

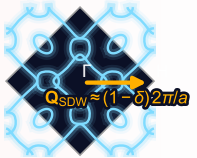
schematic





(b) electrons at E_F

Cr: jack (Γ) nests onto octahedron (H)
⇒ spin- and charge-density waves

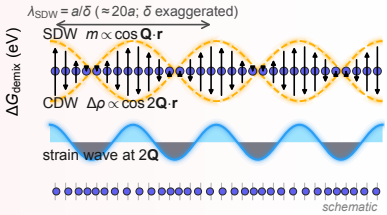


ETT (Lifshitz): adding Cr moves E_F →



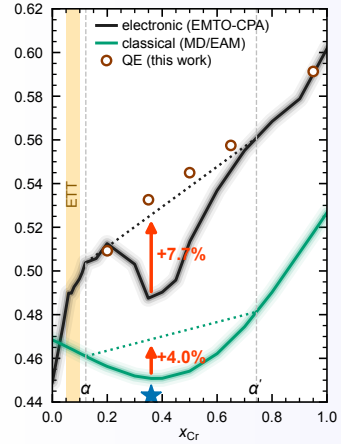
Fe↑ sheet, 5−10% Cr: √-type cusp in *B*, *C*

schematic



schematic

$C = C_{ion} + C_{band}$, $C_{band} \propto -\sum_k (\partial\varepsilon_k/\partial\eta)^2\, \delta(\varepsilon_k - E_F)$



(d) surface acoustic waves

10^4 grains, λ = 6.4 μm; QE at 35% Cr

